

# Neon Drift Courier

## **1. Introduction**

Neon Drift Courier is a fast-paced arcade survival game developed using the C programming language and the raylib graphics library. The objective of the game is to control a courier ship, collect energy orbs to increase the score, and avoid incoming obstacles. The game focuses on smooth movement, visual effects, and responsive gameplay.

This project demonstrates the use of core programming concepts such as loops, functions, structures, and real-time rendering. It also highlights the use of graphical libraries to create an engaging user experience.

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## **2. Game Design**

The game is designed as a continuous survival system where the player must avoid obstacles while collecting points.

### **Key Features:**

- Real-time player movement using keyboard input (W, A, S, D)
- Dynamic obstacle spawning
- Score system based on survival and collection
- Particle effects for visual polish
- Game states (Menu, Playing, Game Over)

The game follows a loop-based architecture where input, update, and rendering happen continuously.

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## **3. How to Play**

- Press **ENTER** to start the game
- Use **W, A, S, D** keys to move the player
- Avoid red obstacles
- Collect glowing orbs to increase score

- If you collide → Game Over
  - Press **ENTER** to restart
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## **4. Implementation Details**

The game is implemented using multiple C source files:

- main.c → Handles initialization and game loop
- game.c → Contains game logic and mechanics
- game.h → Header file for structure and function declarations

### **Key Programming Concepts Used:**

- Structures (for player and objects)
- Functions (modular design)
- Game loop (continuous update system)
- Collision detection
- Random generation for obstacles

Raylib library is used for:

- Graphics rendering
  - Input handling
  - Drawing shapes and animations
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## **5. Flowchart**

Start

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Initialize Game (window, player, variables)

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Main Menu

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[Press ENTER]

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Start Game Loop

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Get Player Input (WASD)

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Update Player Position

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Spawn Obstacles & Orbs

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Check Collisions

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If Collision → Game Over

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Else → Increase Score

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Render Graphics (player, particles, UI)

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Repeat Loop

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Game Over Screen

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[Press ENTER to Restart]

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Back to Game

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## **6. Conclusion**

This project successfully demonstrates the development of a graphical game using C and raylib. The game includes interactive controls, animations, and structured programming. It reflects a clear understanding of programming fundamentals and game design principles.

## **8. GitHub Link**

<https://github.com/aayanr757-web/Neon-Drift-Courier>